TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

Reminders

Standard → slope-intercept: subtract the x-term, then divide every term by the number with y.

Point-slope → slope-intercept: distribute, then add or subtract to get y alone.

Intercepts: x-intercept: put 0 in for y. y-intercept: put 0 in for x.

Solve a formula for a letter: undo everything around that letter, doing the same to both sides.

1. Which equation is $4x + 2y = 10$ written in slope-intercept form?

A. $y = 2x + 5$

B. $y = -2x + 5$

C. $y = -4x + 10$

D. $y = -2x + 10$

Subtract $4x$: $2y = -4x + 10$. Divide by 2: $y = -2x + 5$.

2. What value of x makes $3x - 7 = 11$ true?

F. 4/3

G. 18

H. 6

J. 3

Add 7: $3x = 18$. Divide by 3: $x = 6$.

3. $P = 2\ell + 2w$. What is ℓ when $P = 30$ and $w = 5$?

A. 20

B. 12.5

C. 15

D. **10**

$30 = 2\ell + 10$, so $2\ell = 20$ and $\ell = 10$.

4. $C = 5/9(F - 32)$. What is C when $F = 50$?

F. **10**

G. 18

H. 50

J. −10

$50 - 32 = 18$, and $5/9 \cdot 18 = 10$.